# NULLSPACE

## "See What the Sources Don't Say." — AI-Assisted Counterfactual Schema Alignment & Contradiction Detection Engine


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| **Role & Specialization:** | Principal Software Architect & Lead Full-Stack AI Engineer |
| **System Version & Date:** | Version 1.0.0 Production Release \| November 2026 |
| **Verification Status:** | **100% Automated Pytest & Puppeteer E2E Passed** |


## 1. EXECUTIVE SUMMARY & ARCHITECTURAL MOTIVATION

Modern information retrieval systems and Large Language Model (LLM) search assistants suffer from a fundamental, systemic flaw: **retrieval confirmation bias**. When a decision-maker queries an AI system regarding high-stakes regulatory, fellowship, or procurement guidelines, standard retrieval-augmented generation (RAG) models retrieve documents that discuss *certain topics* and synthesize a fluent summary. However, they fail entirely to answer the counterfactual question: **What is missing from these sources that is mandatory to make an informed decision?**

**NullSpace** inverts this paradigm through **Counterfactual Ontology Generation**. Before fetching or reading any external URLs, the engine synthesizes an independent Decision Requirement Graph $G = (V, E)$. When ingested source documents fail to provide grounded claims for a mandatory requirement, the system mathematically flags a **Digital Dark Spot** ($InDegree(v) = 0$). Furthermore, pairwise natural language inference compares claims mapped to identical criteria to expose subtle numeric divergences and policy contradictions that traditional summarizers blend into false harmony.

## 2. MATHEMATICAL FORMULATIONS & EPISTEMIC PRINCIPLES

**A. Digital Dark Spot Condition:**
Let $V_{\text{mandatory}} \subseteq V$ be the set of mandatory decision nodes synthesized in the counterfactual schema. For any node $v \in V_{\text{mandatory}}$, the Digital Dark Spot indicator is defined strictly by the directed in-degree of grounded claim mappings:

$$\textbf{IsDarkSpot}(v) \Leftrightarrow \text{InDegree}(v) == 0, \quad \text{where InDegree}(v) = |\{ c \in \text{Claims} : \text{mapped}(c, v) \}|$$

This formulation guarantees that unaddressed decision facets are mathematically surfaced rather than hallucinated or omitted.

**B. Immutable Character Span Grounding:**
To prevent LLM paraphrase hallucinations from being cited as source facts, NullSpace preserves raw document snapshots with cryptographic hashing:

$$\text{ContentHash} = \text{SHA256}(\text{RawText})$$

Every claim extracted must satisfy the exact substring identity: **RawText[char_start : char_end] == exact_quote**. If a model hallucinates a paraphrase, the offset verification fails, and the claim is barred from the Grounded Evidence Vault.

**C. Normalized Shannon Evidence Entropy:**
For a requirement node $v$ with $N > 0$ claims classified across $k=4$ discrete epistemic states (Support, Contradiction, Stale, Unverified):

$$H(X_v) = -\sum p_i \log_2(p_i), \qquad H_{norm}(X_v) = H(X_v) / \log_2(k)$$

When $N = 0$ (Dark Spot), missingness uncertainty is defined as maximum: **U_missing = 1.0**.

**D. Priority Risk Scoring Formula (0 - 100):**
PriorityScore(v) = min(100, [0.35 × (Imp/5) × 35 + 0.30 × U_missing × 30 + 0.20 × I_conflict × 20 + 0.15 × I_stale × 15 + 0.10 × H_norm × 10])
• Critical Risk (≥ 75) | • High Risk (≥ 55) | • Medium Risk (≥ 35) | • Low Risk (< 35)

## 3. SYNTHETIC CASE STUDY AUDIT: NATIONAL PRERANA MERIT FELLOWSHIP

To prove the engine's capabilities without external network dependencies, NullSpace incorporates a deterministic synthetic fixture with three conflicting documents:

| Requirement Node | Gazette Notice (2026-27) | Campus Pulse (AY 2025-26) | Education Watch (2024) | Engine Finding |
|---|---|---|---|---|
| Income Threshold | Rs. 3,00,000 ceiling | Rs. 4,00,000 expanded | Rs. 2,00,000 limit | CRITICAL CONFLICT (3-Way) |
| Renewal Policy | Deferred / Omitted | Mandatory with 7.5 CGPA | Renewal Not Required | PRIMARY CONTRADICTION |
| Required Documents | Marksheet & Income Cert | Marksheet & Income Cert | Bank Statement & Income | DOCUMENT DISCREPANCY |
| Application Portal | Not Mentioned | Not Mentioned | Not Mentioned | DIGITAL DARK SPOT (InDegree=0) |

## 4. SYSTEM VERIFICATION & AUTOMATED TESTING TELEMETRY

A rigorous multi-tier testing regimen was executed across backend, frontend, and browser integration boundaries:

| Test Suite Component | Scope & Target | Checkpoints | Status |
|---|---|---|---|
| Pytest Epistemic Unit Suite | Ontology, SSRF, Offsets, NLI, Entropy | 13 Checkpoints | PASSED (100%) |
| Pytest API Integration Suite | Health, Demo Load, Exports, Workflow | 3 Multi-stage flows | PASSED (100%) |
| Frontend TypeScript Engine | Strict compilation & interfaces | tsc --noEmit | PASSED (0 Errors) |
| Vite Production Bundler | Static tree-shaking & CSS build | 1,555 modules | PASSED (9.35s) |
| Chromium E2E Automation | All 9 tabs, modal, dorks, exports | 4 Visual journeys | PASSED (Screenshots verified) |

## 5. SECURITY & THREAT MITIGATION LAYER

**Server-Side Request Forgery (SSRF) Protection:**
NullSpace implements strict IP address and protocol validation. Before any HTTP socket is opened, the hostname is resolved via DNS and checked against RFC 1918 private ranges, loopback (127.0.0.0/8, ::1), carrier-grade NAT, and cloud metadata endpoints (169.254.169.254). Redirects are limited to 3 hops and re-validated at each step.
**Prompt Injection Immunity:**
Source document text is strictly treated as untrusted data. It is never concatenated into system prompt instructions. Extracted claims are grounded via exact character offsets rather than unconstrained generative re-writing.

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